

PDF 2

SUPERVISED MACHINE LEARNING MODELS FOR WEB SERVICE CLASSIFICATION

(Chipa et al., 2019)

TITLE PAGE

Supervised Machine Learning Models for Web Service Classification

Mini Project Report

Submitted By:

- Kuresu Sahu
- Mahes Kumar Sahu
- Sourava Pradhan
- Srikanta Panigrahy

Department of Computer Science and Engineering

Roland Institute of Technology

2026

ABSTRACT

Supervised machine learning algorithms are widely used for classification tasks. This report studies different supervised learning models used for web service classification, including Decision Tree, Support Vector Machine, K-Nearest Neighbor, and Logistic Regression.

The report evaluates classification performance, advantages, disadvantages, and limitations of supervised learning methods in web service quality prediction systems.

INTRODUCTION

Supervised learning is a machine learning technique where models are trained using labeled datasets.

In web service classification, the model learns patterns from existing services and predicts quality categories for unknown services.

Common supervised algorithms include:

- KNN
- SVM
- Decision Tree
- Logistic Regression

OBJECTIVES

1. To study supervised learning algorithms.
2. To classify web services.
3. To compare algorithm performance.
4. To improve prediction accuracy.
5. To analyze classifier limitations.

EXISTING PROBLEM

Problems in Existing Systems

- Large labeled dataset requirement
- High manual labeling cost
- Poor performance on unseen data
- Overfitting issues
- Low adaptability

K-NEAREST NEIGHBOR (KNN)

KNN classifies data based on nearest neighboring samples.

Advantages

- Simple implementation
- Good for small datasets

Disadvantages

- Slow for large datasets
- Sensitive to noise

SUPPORT VECTOR MACHINE (SVM)

SVM separates classes using hyperplanes.

Advantages

- High-dimensional handling
- Better classification accuracy

Disadvantages

- Complex tuning
- Slow training process

DECISION TREE

Decision Tree creates classification rules using branching structures.

Advantages

- Easy interpretation
- Fast prediction

Disadvantages

- Overfitting
- Low generalization

LOGISTIC REGRESSION

Logistic Regression is used for linear classification problems.

Advantages

- Fast execution
- Simple implementation

Disadvantages

- Weak nonlinear handling

METHODOLOGY

1. Data Collection
2. Data Preprocessing
3. Feature Extraction
4. Training
5. Prediction
6. Evaluation

OBSERVATION

Decision Trees performed better in interpretability while SVM achieved better separation performance. However, all supervised models struggled with limited labeled data.

RESULT ANALYSIS

Algorithm	Accuracy
KNN	45%
Decision Tree	38%
SVM	24%
Logistic Regression	56%

APPLICATIONS

- Recommendation systems
- E-commerce
- Healthcare prediction
- Service ranking systems

CONCLUSION

Supervised machine learning algorithms provide efficient classification performance but heavily depend on labeled datasets. Semi-supervised methods may provide better scalability and accuracy.

BIBLIOGRAPHY

1. Chipa et al., "Supervised Machine Learning Models", 2019.
2. Scikit-learn Documentation.
3. IEEE Machine Learning Research Papers.